Caden Tebow

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Education

The University of Chicago Chicago, IL
Bachelor of Science in Physics and Computer Science (GPA: 3.94/4.00) Expected June 2028
Relevant Coursework: DSA (DPV), Differential Equations, Systems Programming, Quantum Mechanics I-II (Griffiths)
Awards: Dean's List 2024-2026, U.S. Presidential Scholar Semi-Finalist
Activities: Student Tour Guide, Kappa Theta Pi (Technology Fraternity), PULSE-A, Chamber Orchestra, SPS

Skills

Programming Languages: Python, C, C++, Assembly
Scientific Libraries: NumPy, SciPy, GEANT-4, ROOT, rooUnfold, OpenCV, MatPlotLib, PyTorch,
Developer Tools: Git, Linux, Docker, Valgrind, GDB, Criterion, Firmware Development (Teensy, Arduino)

Experience

NASA Goddard Space Flight Center - Energetic Particle Laboratory Greenbelt, MD
Student Researcher Intern June 2026 - August 2026

- Simulate single and double-scatter neutron spectrometers using GEANT-4 to model the effects of scintillator quenching and SiPM quantum efficiency on detector energy resolution.
- Develope a custom unfolding pipeline using rooUnfold to execute both regularized (iterative bayesian unfolding) and un-regularized (matrix inversion) unfolding methods to recover double-scatter energy spectra of energies through 100 MeV.

Kavli Institute for Cosmological Physics - UChicago Vieregg Lab Chicago, IL
Undergraduate Researcher January 2025 – Present

- Engineer custom firmware for radar sensors, integrating control via a Teensy microcontroller and Python automation scripts, validating the performance of radar sensors and creating faster testing workflows while achieving 1mm accuracy.
- Design end-to-end custom PCB hardware, creating a KiCad-based breakout board for a precision distance sensor, selecting components, building full schematics, and running electrical simulations and design checks.

PULSE-A CubeSat - NASA Undergraduate Satellite Chicago, IL
Payload Engineer September 2025 – Present

- Develop finite-state-machine spiral-search algorithms for a 3U CubeSat's acquisition system, enabling autonomous target detection for a 2027 mission demonstrating laser communications via circular-polarized shift-keying (CPolSK).
- Benchmark, test, and deploy a MEMS Fine Steering Mirror (FSM) with custom C++ firmware to ∼3% error from documented benchmarks while in an ESD and cleanroom setting.

Selected Presentations and Publications

- **Tebow, C.**, De Nolfo, G., Pudajas, R., & Mitchell, J.G. (2026). "Evaluating the Effectiveness of Single-Scatter Neutron Spectrometers and the nDURE Instrument for Detecting High Energy Albedo Neutrons" *2026 NASA Goddard Student Research Symposium*, https://ugradresearchsymposium2026.omeka.net/items/show/672
- **Tebow, C.**, Schulze-Kalt, G., Marsden, C., & Zhong, T. (2026). "MEMS Fine-Steering Mirror Characterization and Control for PULSE-A Free-Space Optical Communication" *2026 UChicago Undergraduate Research Symposium*, https://ugradresearchsymposium2026.omeka.net/items/show/672

Selected Projects

Drawn Circuit Identifier: github.com/ctebow/breadMaker-backend June 2025

- Developed a machine learning model that was trained on 10,000+ images using PyTorch and OpenCV for automatic hand-drawn circuit element uploads, streamlining the process to be 2x faster.
- Tuned model over multiple epochs using a local GPU and manually bounded classes, with validation on a 300-image test set achieving an 85% mAP@50 with >80% confidence across 30+ circuit classes.

Retrieval-Augmented Generation Pipeline: github.com/ctebow/Local-RAG-with-Gemini August 2025

- Engineered a retrieval-augmented generation pipeline using vector embeddings with chunking and top-k search, enabling for automatic context generation when prompting an LLM.
- Optimized for constrained local hardware to support 2,000+ token responses within a 4,000-token context window, allowing for <20 second response times and saving 15 seconds per LLM prompt.